

JMeter Load Test Report: Login and Profile API

1 Test Plan Overview

The performance test was designed and executed in Apache JMeter to evaluate the behavior of the Login API and the Profile API under varying levels of concurrent load. The test plan consisted of the following components.

1.1 CSV Data Set Config

A CSV Data Set Config element was used to parameterize login credentials. The CSV file contained 10 entries of `email,password` pairs, which were used as variables in the Login API request body. One entry in the CSV file was deliberately configured with an incorrect email/password combination in order to introduce a controlled variation (i.e., a login failure) into the test data and observe its downstream effect on the Profile API calls.

1.2 HTTP Requests

Login API (POST)

- **Endpoint:** `https://devapi.remoteaxle.com/login`
- **HTTP Headers:**
 - `Content-Type: application/json; charset=utf-8`
 - `Accept: application/json, text/plain, */*`
- **Body:** email and password variables sourced from the CSV Data Set Config.

JSON Extractor (on Login API response)

- **Variable Name:** `token`
- **JSON Path Expression:** `$.data.access_token`
- **Match No.:** `1`
- **Default Value:** `NOT_FOUND`

Profile API (GET)

- **Endpoint:** `https://devapi.remoteaxle.com/users/profile`
- **HTTP Headers:**
 - `Authorization: Bearer ${token}`

The extracted `token` from the Login API response was injected into the `Authorization` header of the subsequent Profile API request, chaining the two requests per iteration.

1.3 Timer

A Uniform Random Timer was applied between requests, configured with:

- **Random Delay Maximum:** 1000 ms (the upper bound from which a random delay value is drawn for each request)
- **Constant Delay Offset:** 2000 ms

The effective delay for each request follows the formula:

$$\text{Delay} = (0.1 \times \text{Random Delay}) + \text{Constant Delay Offset}$$

where *Random Delay* is a value drawn uniformly at random between 0 ms and the configured Random Delay Maximum (1000 ms) for that specific request.

1.4 Assertions

The following assertions were applied to each sampler (Login API and Profile API):

- **Response Code Assertion**
- **Response Assertion** (message-based validation)
- **Size Assertion**

1.5 Listeners

Results were captured using the **Summary Report** and **Aggregate Report** listeners.

2 Test Execution Scenarios

Five load scenarios were executed by varying the Thread Group configuration (Number of Threads, Ramp-Up Period, Loop Count). Since the CSV Data Set Config recycles through its 10 credential entries, the total number of samples per sampler equals Threads $\times$ Loop Count, regardless of the Ramp-Up Period used. Scenarios below are ordered by increasing load: primarily by thread count (concurrency), and by total sample volume as a tiebreaker where thread count is equal. For brevity, each report below is limited to the core metrics: Samples, Average, Min, Max, and Error % for the Summary Report; and Samples, Average, Median, 90% Line, 95% Line, and 99% Line, and Error % for the Aggregate Report.

Scenario 1: Threads = 1, Ramp-Up = 1s, Loop Count = 1

Summary Report

Label	#Samples	Average	Min	Max	Error %
Login API CSV	1	754	754	754	0.00%
Profile API	1	212	212	212	0.00%
TOTAL	2	483	212	754	0.00%

Aggregate Report

Label	#Samples	Average	Median	90% Line	95% Line	99% Line	Error %
Login API CSV	1	754	754	754	754	754	0.00%
Profile API	1	212	212	212	212	212	0.00%
TOTAL	2	483	754	754	754	754	0.00%

Scenario 2: Threads = 10, Ramp-Up = 10s, Loop Count = 10

Summary Report

Label	#Samples	Average	Min	Max	Error %
Login API CSV	100	580	187	859	10.00%
Profile API	100	169	18	482	10.00%
TOTAL	200	374	18	859	10.00%

Aggregate Report

Label	#Samples	Average	Median	90% Line	95% Line	99% Line	Error %
Login API CSV	100	580	589	737	744	839	10.00%
Profile API	100	169	129	297	359	398	10.00%
TOTAL	200	374	346	715	737	821	10.00%

Scenario 3: Threads = 100, Ramp-Up = 20s, Loop Count = 5

Summary Report

Label	#Samples	Average	Min	Max	Error %
Login API CSV	500	681	165	1361	10.00%
Profile API	500	69	10	233	93.40%
TOTAL	1000	375	10	1361	51.70%

Aggregate Report

Label	#Samples	Average	Median	90% Line	95% Line	99% Line	Error %
Login API CSV	500	681	590	1127	1212	1312	10.00%
Profile API	500	69	69	82	116	156	93.40%
TOTAL	1000	375	168	1018	1127	1291	51.70%

Scenario 4: Threads = 200, Ramp-Up = 20s, Loop Count = 10

Summary Report

Label	#Samples	Average	Min	Max	Error %
Login API CSV	2000	3930	166	9389	10.00%
Profile API	2000	74	10	2716	91.35%
TOTAL	4000	2002	10	9389	50.68%

Aggregate Report

Label	#Samples	Average	Median	90% Line	95% Line	99% Line	Error %
Login API CSV	2000	3930	4707	5621	5726	5905	10.00%
Profile API	2000	74	71	110	124	165	91.35%
TOTAL	4000	2002	227	5446	5621	5837	50.68%

Scenario 5: Threads = 100, Ramp-Up = 100s, Loop Count = 100

Summary Report

Label	#Samples	Average	Min	Max	Error %
Login API CSV	10000	928	159	8405	10.00%
Profile API	10000	160	11	6952	93.32%
TOTAL	20000	544	11	8405	51.66%

Aggregate Report

Label	#Samples	Average	Median	90% Line	95% Line	99% Line	Error %
Login API CSV	10000	928	818	1424	1624	2483	10.00%
Profile API	10000	160	93	236	320	1159	93.32%
TOTAL	20000	544	323	1217	1444	2178	51.66%

3 Observations

- Baseline (1 thread): Login 754 ms, Profile 212 ms, 0% error.
- From Scenario 2 onward, Login API error is consistently 10% — matches the single bad CSV row (1 in 10).
- Profile API error is far higher in all concurrent scenarios: 93.40% (Scenario 3), 91.35% (Scenario 4), 93.32% (Scenario 5) — indicating token extraction or authorization issues under load, not just bad credentials.
- Response times degrade with thread count: Login average rises from 754 ms (Scenario 1) to 3930 ms (Scenario 4, 200 threads), with 99th percentile reaching 5905 ms and max hitting 9389 ms.
- The 100-thread / 100-loop test (Scenario 5) shows moderate degradation (avg 928 ms) due to slower ramp-up (100s), confirming that ramp-up rate significantly affects server strain.

Key Finding: Token/Authorization Bottleneck at High Concurrency

Across Scenarios 3–5, Profile API error rate (91–93%) vastly exceeds Login API error (10%), proving the failure is not due to bad credentials. At 200 threads (Scenario 4), Login response times spike severely (99th %ile: 5905 ms, max: 9389 ms), indicating the server struggles with both token handling and raw throughput. The slower ramp-up in Scenario 5 (100s) mitigates peak load, suggesting that gradual ramp-up reduces contention.